



NCV LEVEL 2: MANUAL MANUFACTURING EXAM BLUEPRINT

The 8-Hour High-Efficiency Study Strategy



Exam Focus: Topics 1 & 3 (90% Total Weight)

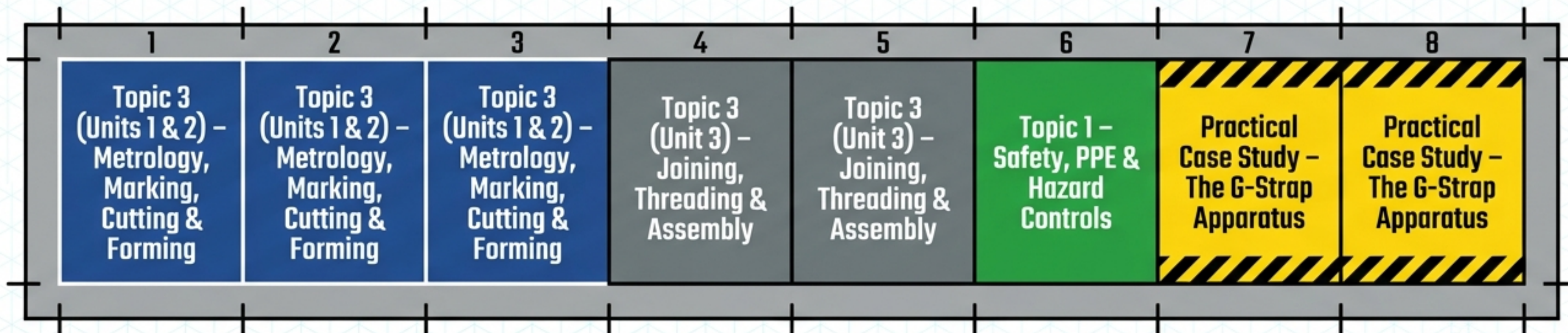


High Yield: Foundation

High Yield: Mechanical

Easy Marks

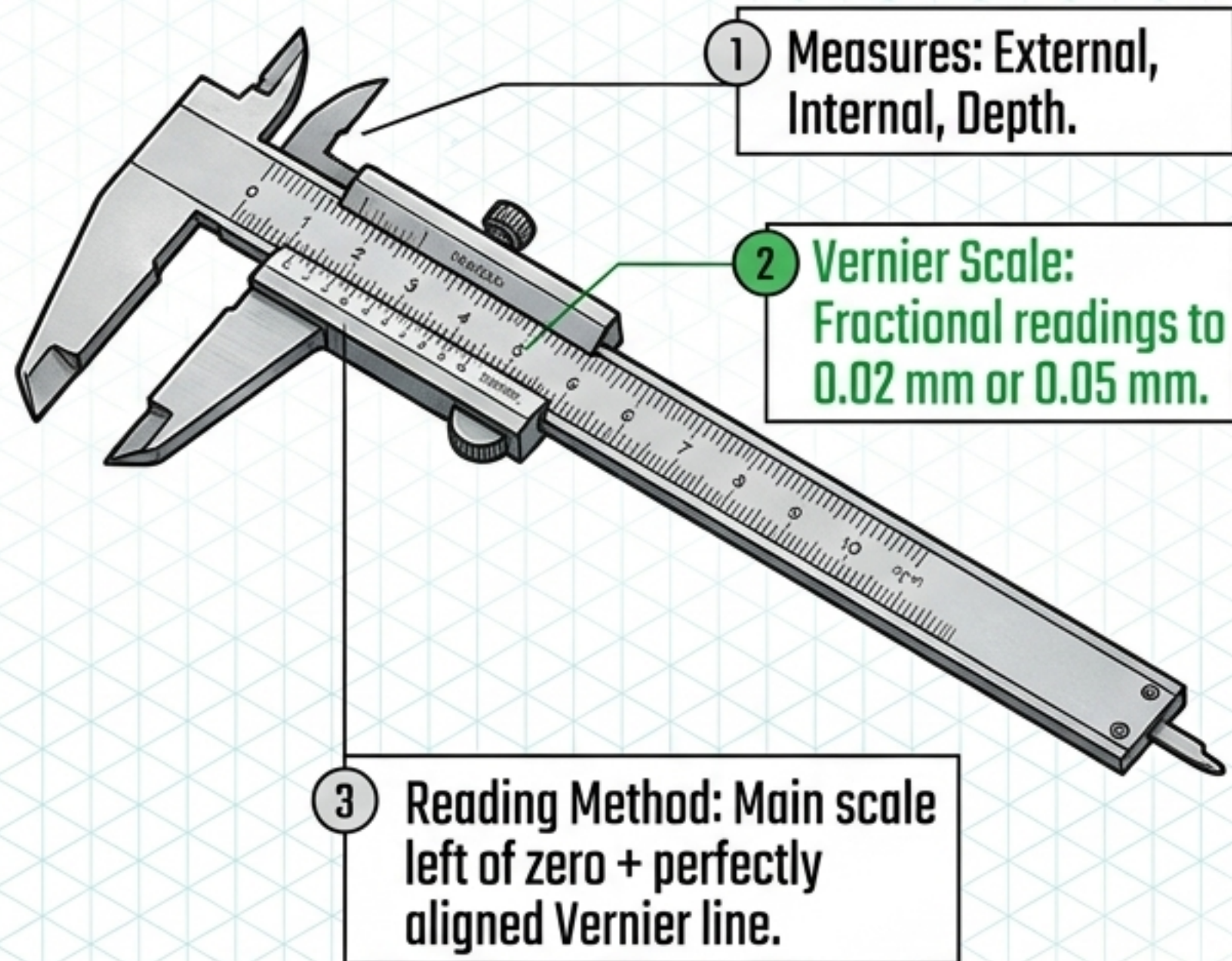
Master Synthesis / 80% Weight Target



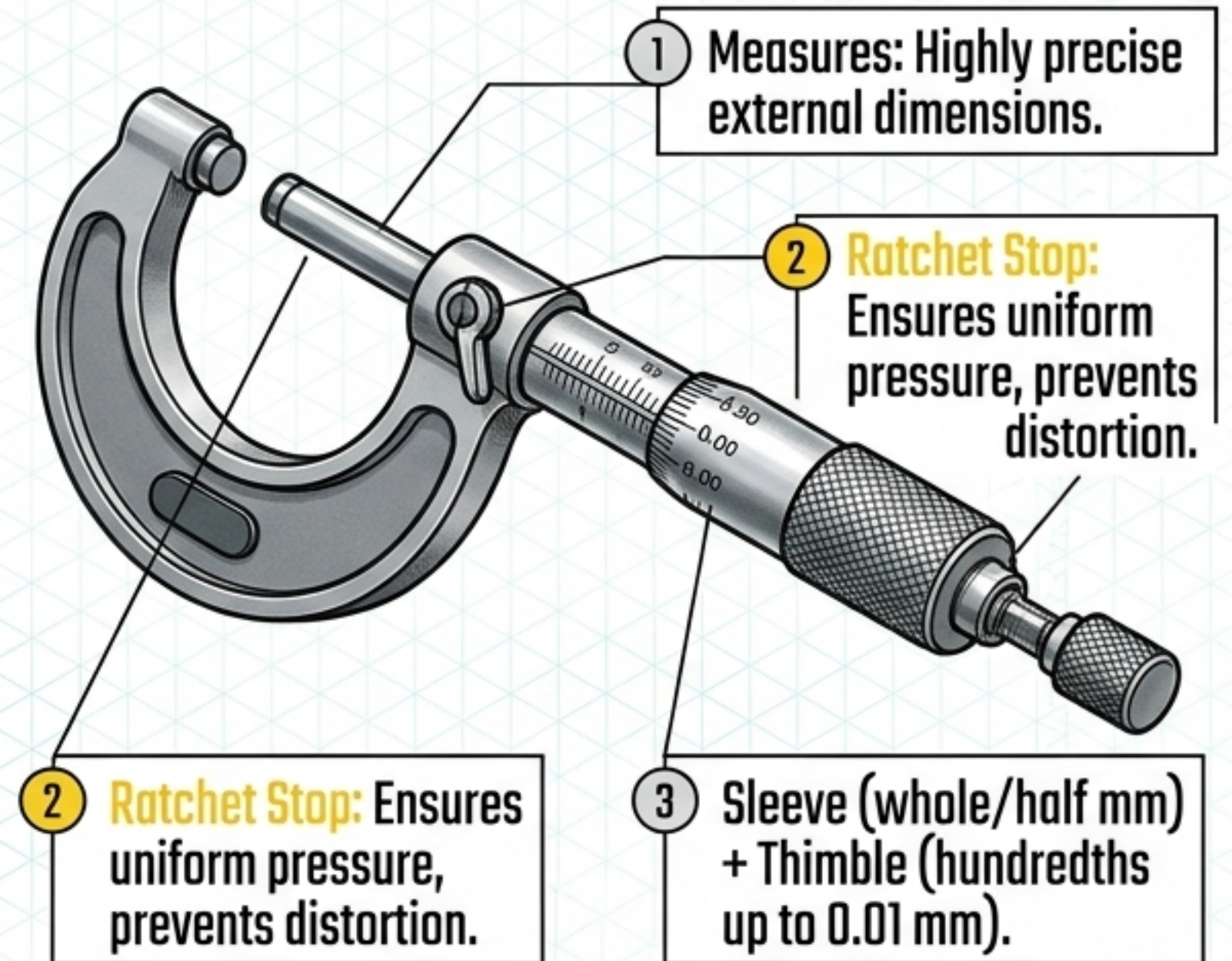
**Topic 3 carries 80% of the exam marks.
The G-strap practical is now a theoretical
assessment.
Master the workflow to pass.**

The Metrology Matrix

Vernier Caliper



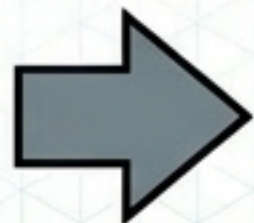
Outside Micrometer



Marking Out Flow

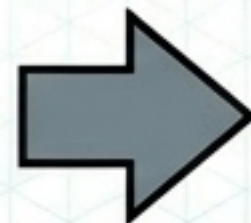
Step 1: Layout Dye

Engineers' Blue applied for visibility.



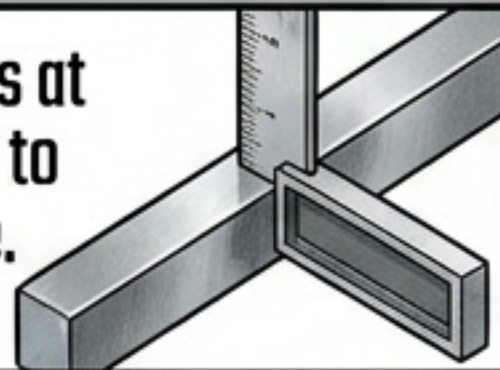
Step 2: Scriber

Tungsten-carbide scratches precise lines.



Step 3: Engineers' Try Square

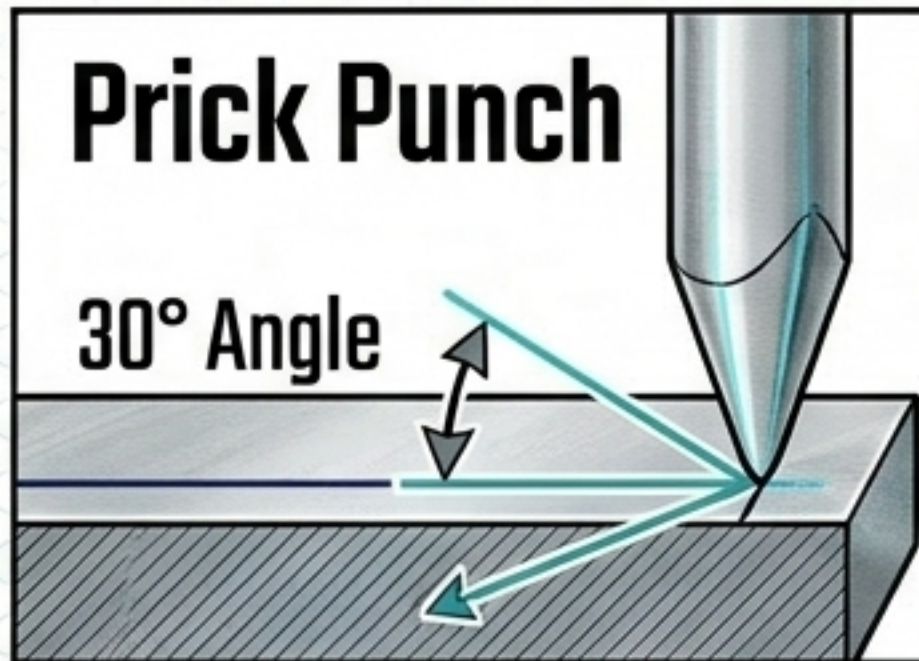
Checks lines at exactly 90° to datum edge.



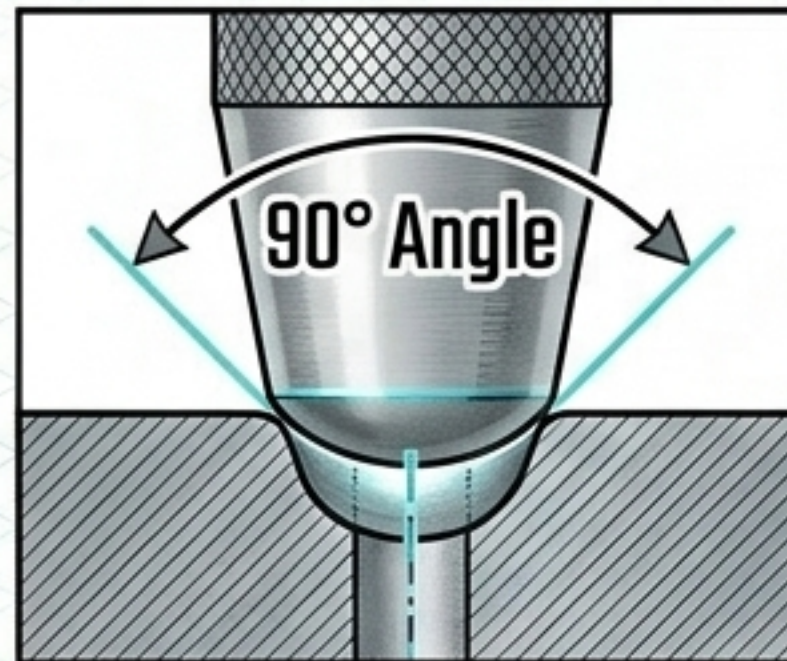
The Punch Selection Matrix

Prick Punch

30° Angle



Function: Small indentations for permanent visibility along scribed lines.



Center Punch

Function: Deep indentation at hole center.

! Critical Metric:
Prevents drill walk.

The TPI Snag Graphic & Blade Selection

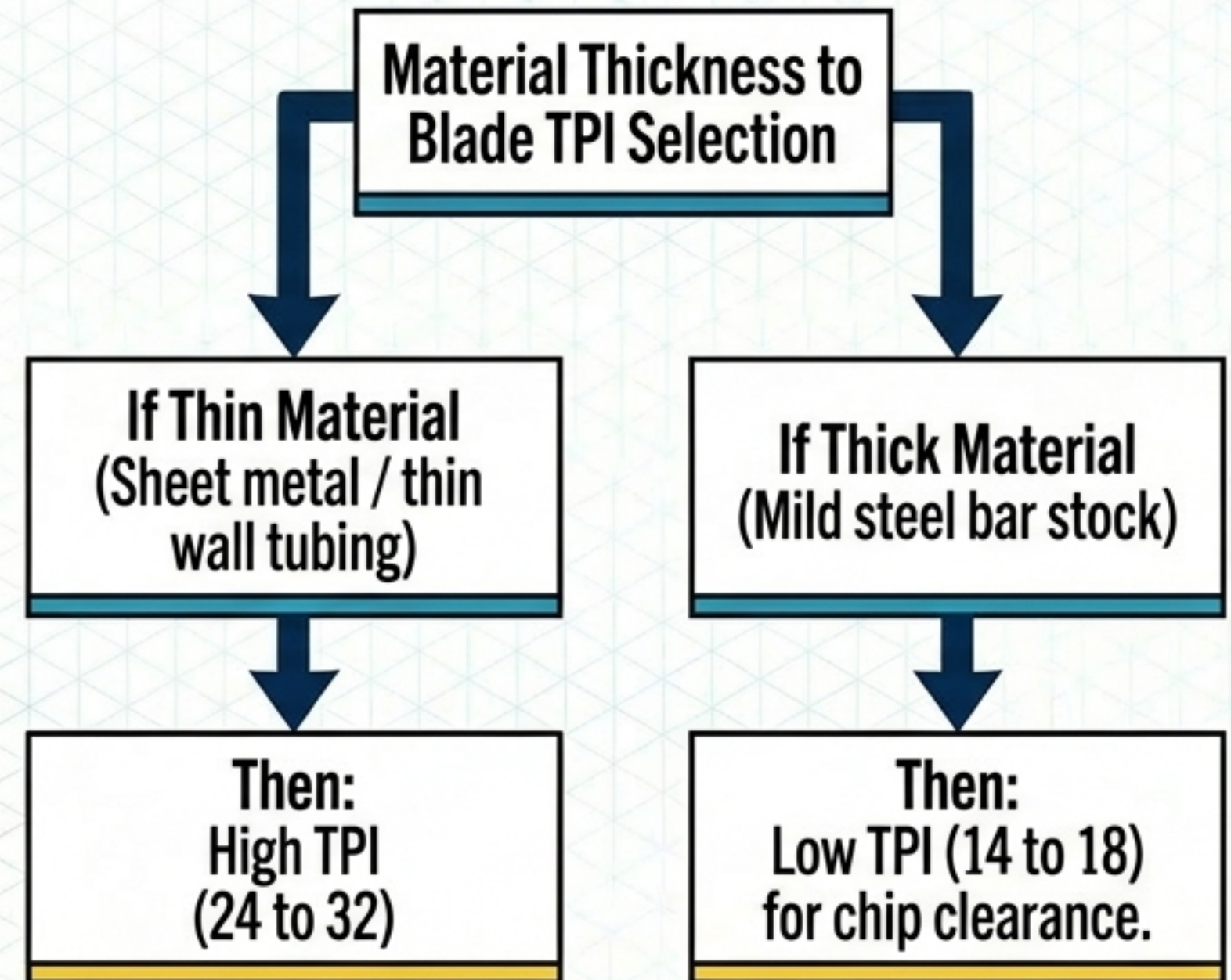
The TPI Snag Graphic



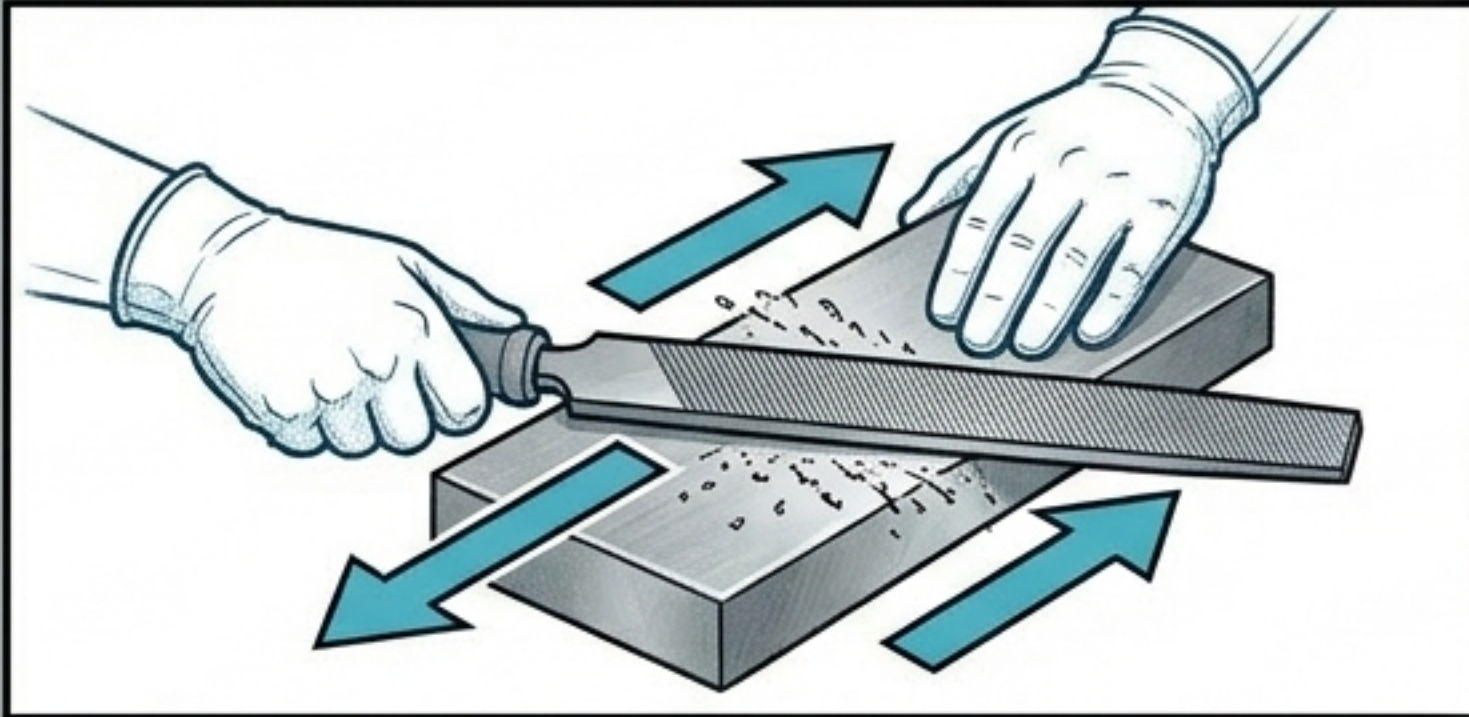
Minimum 2 to 3 teeth must contact the cross-section at all times to prevent snagging and stripping.

- ✓ Teeth point forward.
- ✓ Pressure on forward stroke only.
- ✓ Release on return.

Blade Selection Logic

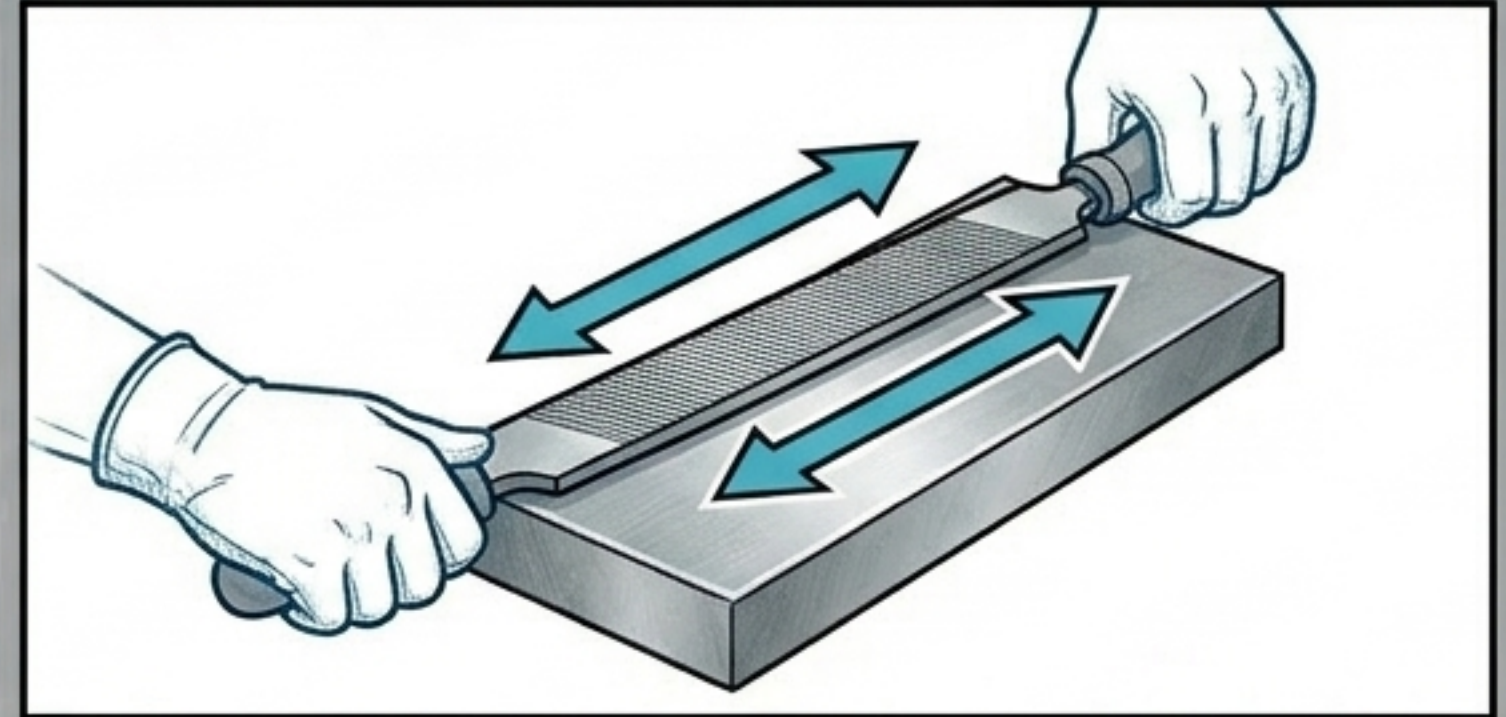


Cross-Filing



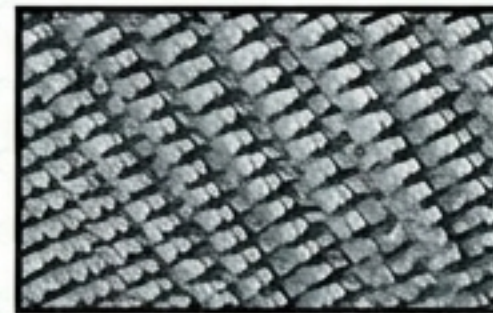
Purpose: Rapid material removal and flattening.

Draw-Filing



Purpose: Smooth, fine surface finish.

The File Grade Matrix



Bastard
Coarse material removal.



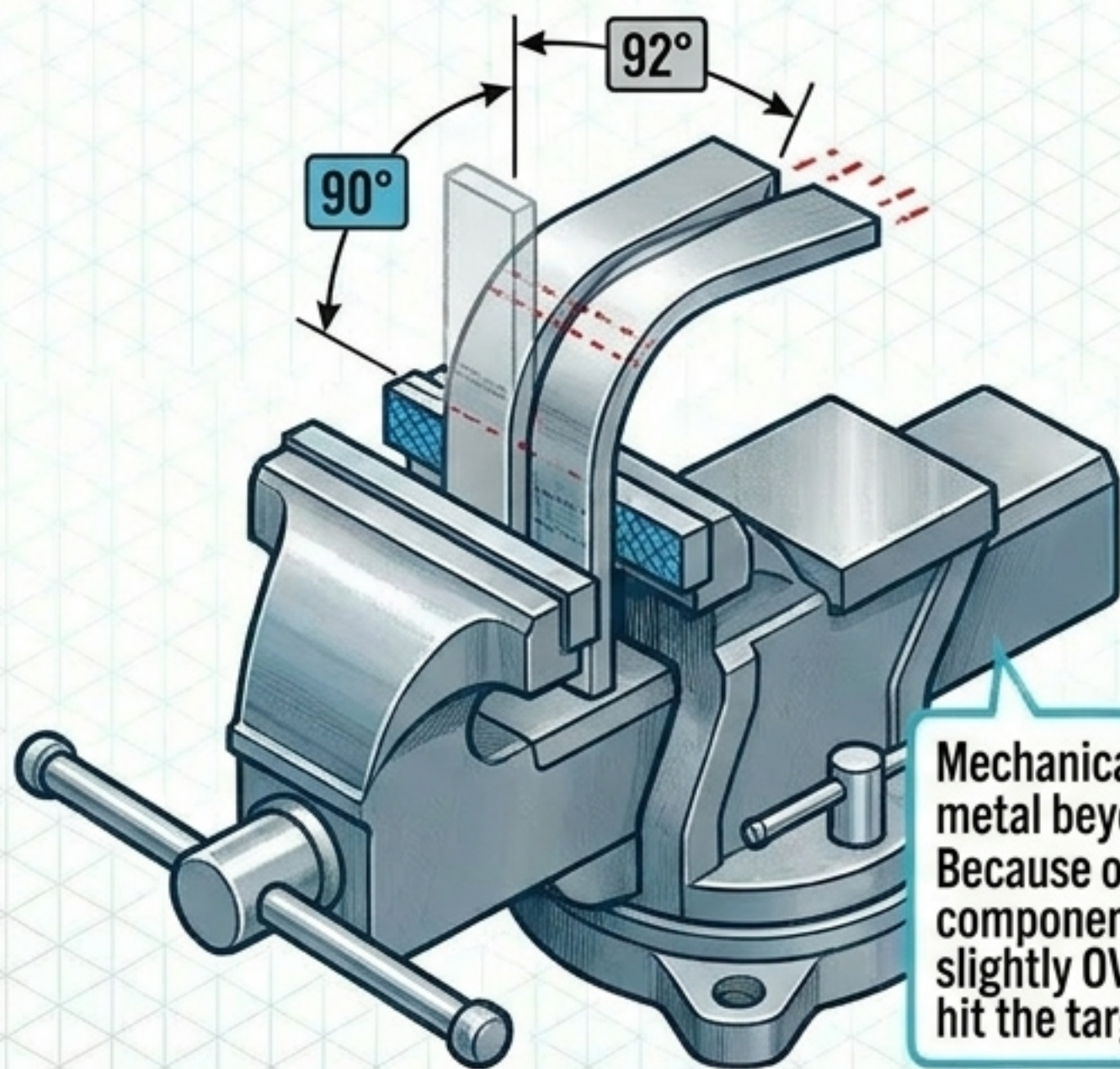
Second-Cut
Medium shaping.



Smooth
Final finishing.

THE SPRINGBACK MODEL & NEUTRAL AXIS PHYSICS

THE SPRINGBACK MODEL



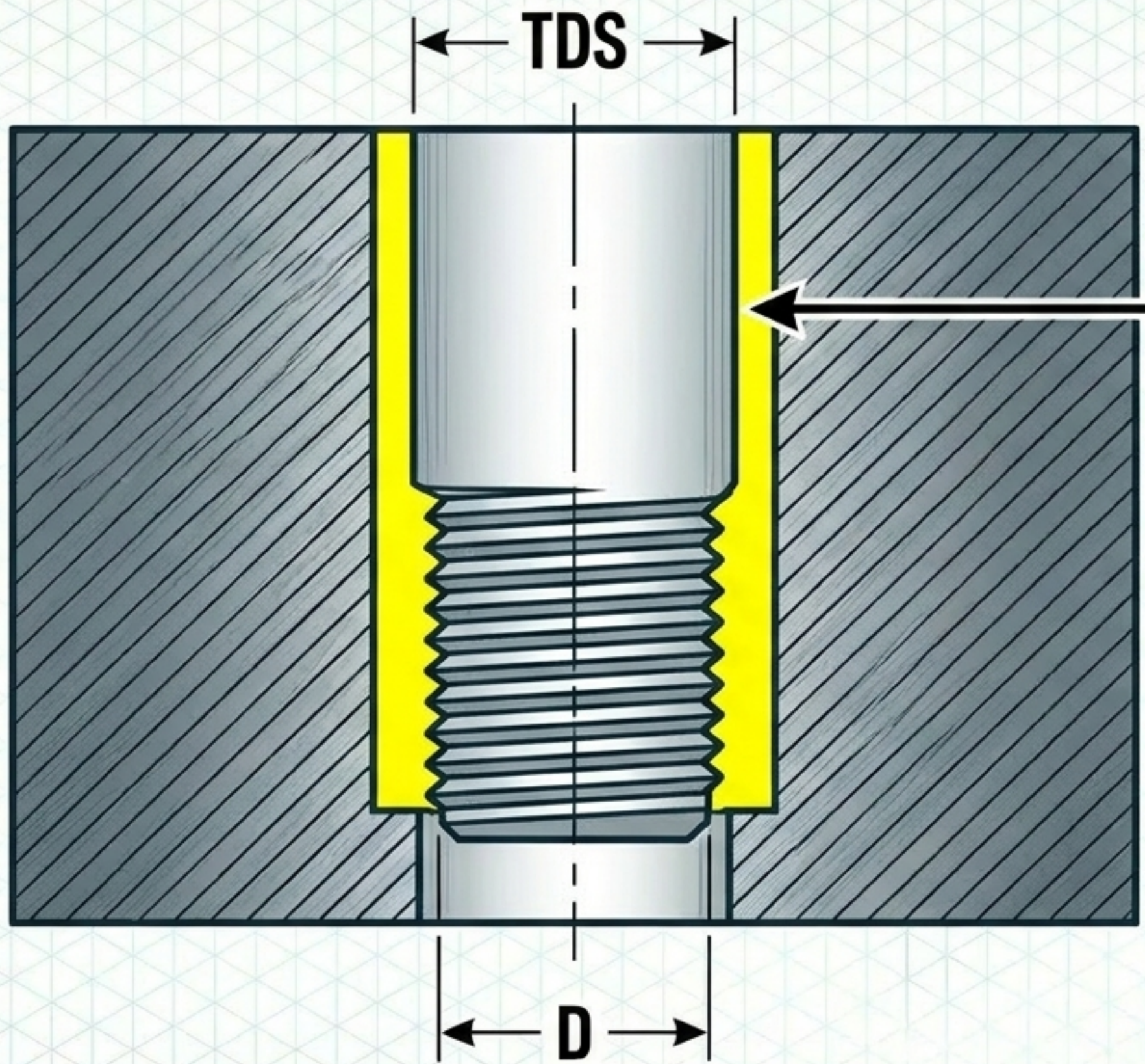
Mechanical Bending forces metal beyond yield point. Because of elasticity, component must be slightly OVER-BENT to hit the target angle.

NEUTRAL AXIS PHYSICS

Expansion
(Tension)

Compression

The Neutral Axis
(Zone of zero expansion/compression used to calculate Bend Allowance)

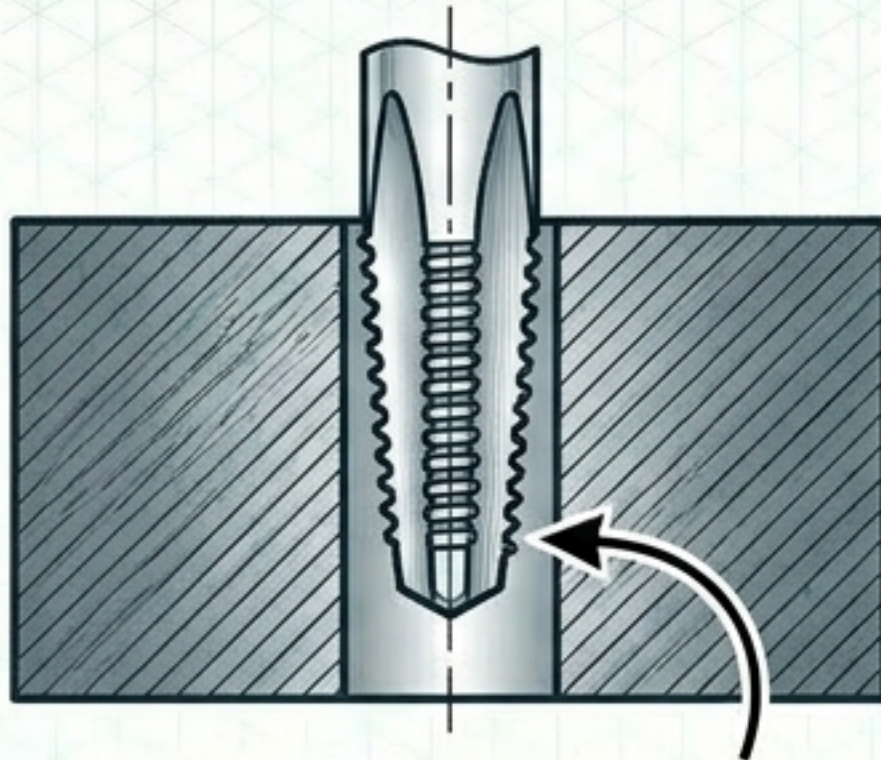


The hole must be smaller than the bolt's nominal diameter so material remains for the tap to cut into.

$$TDS = D - P$$

(Tap Drill Size = Nominal Diameter - Thread Pitch)

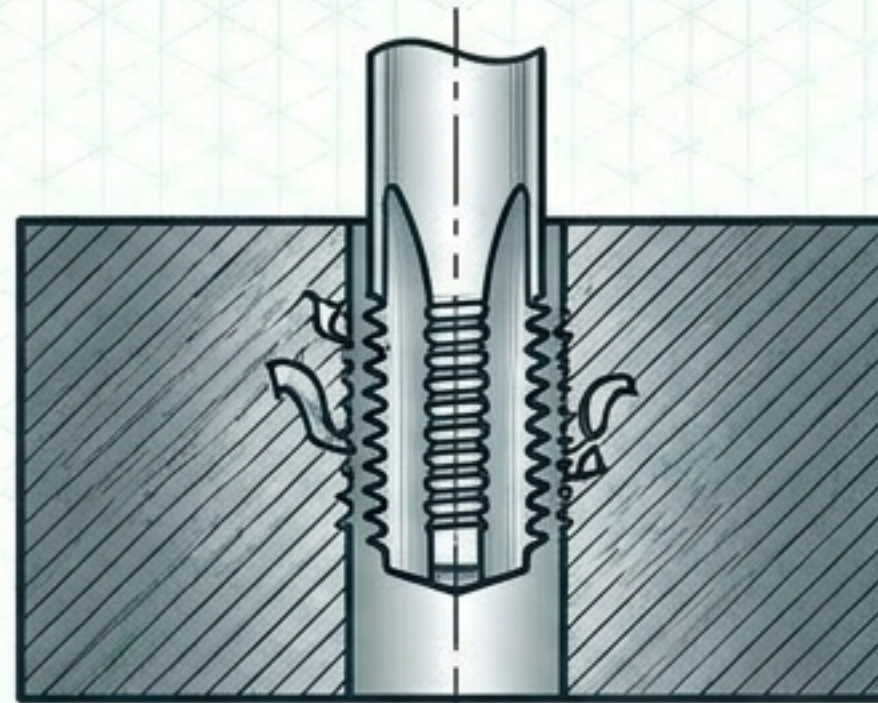
TAPER TAP (FIRST TAP)



8-10 threads tapered.

Function: Easily aligns and starts the thread square.

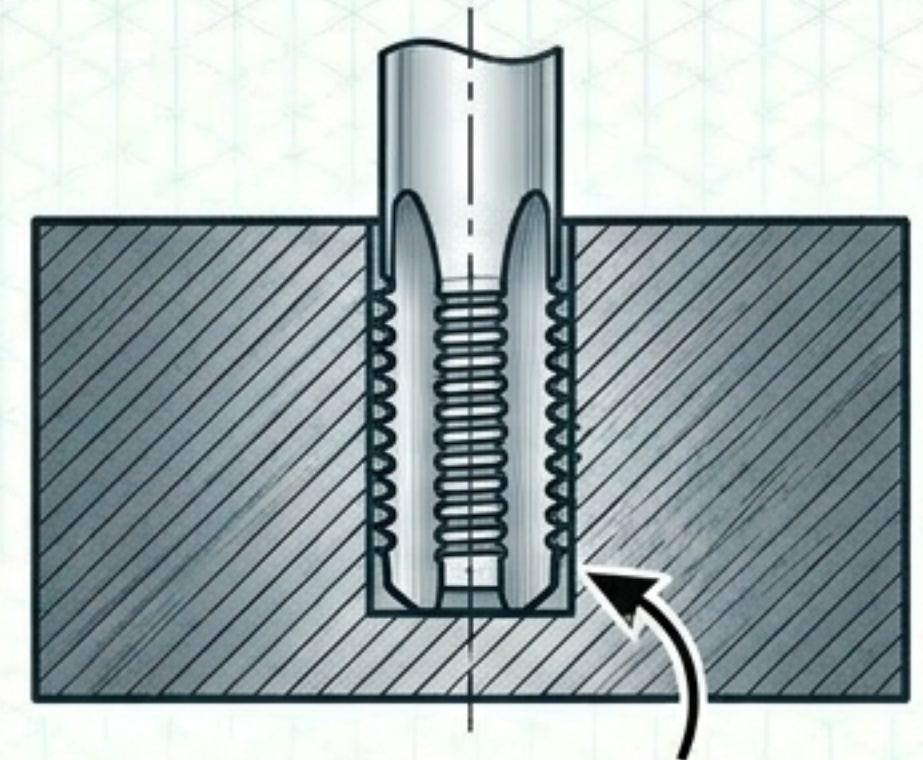
INTERMEDIATE / SECOND TAP



3-5 threads tapered.

Function: Deepens the started threads.

BOTTOMING / PLUG TAP (THIRD TAP)



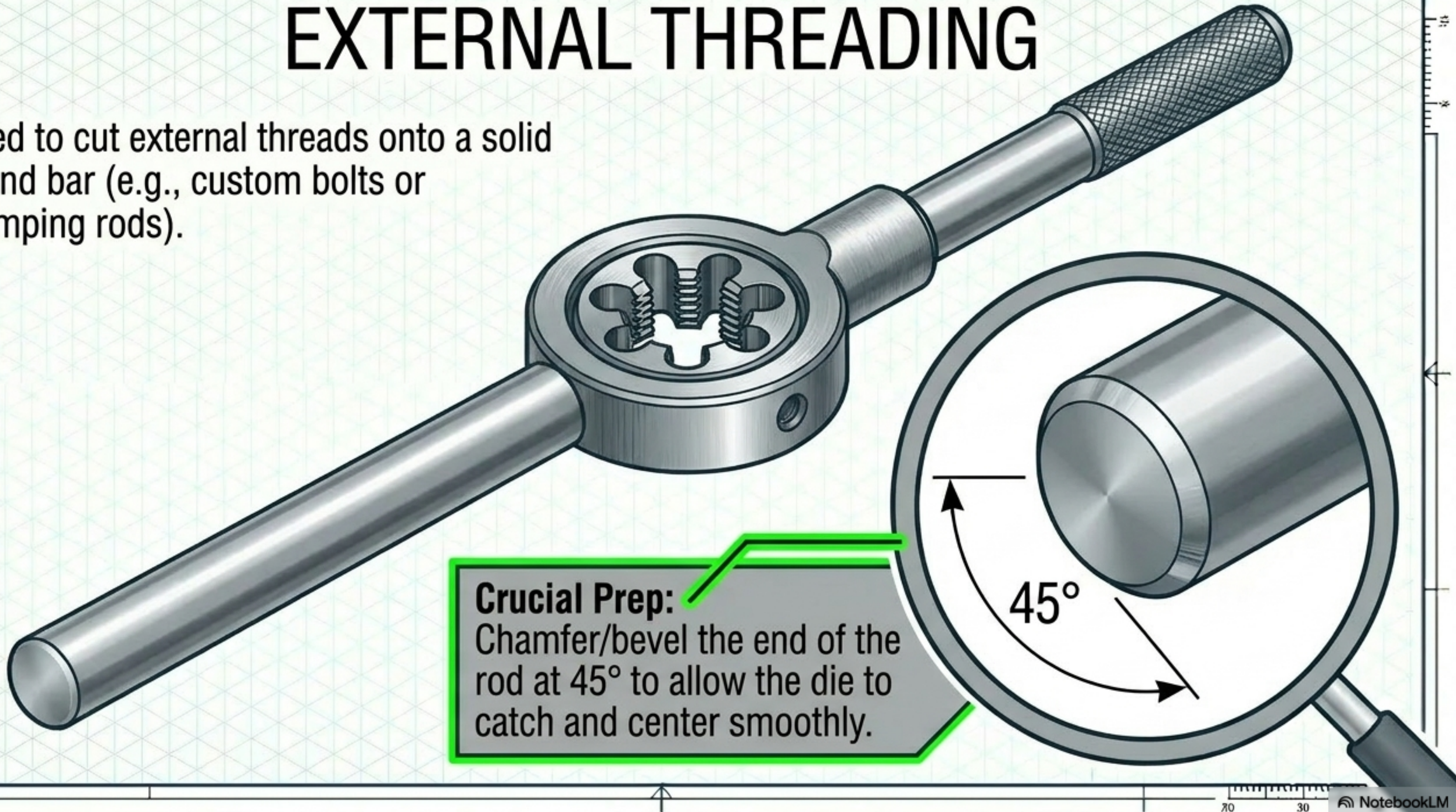
1-2 threads tapered.

Function: Cuts full-depth threads all the way to the bottom of a blind hole.

Operational Technique: Turn wrench 1/2 to 1 full turn clockwise, then reverse 1/4 to 1/2 turn to **BREAK THE CHIP**. Use cutting fluid.

EXTERNAL THREADING

Used to cut external threads onto a solid round bar (e.g., custom bolts or clamping rods).



Crucial Prep:

Chamfer/bevel the end of the rod at 45° to allow the die to catch and center smoothly.

PERSONAL PROTECTIVE EQUIPMENT (PPE) MATRIX



Drilling / Grinding

Hazard: High-velocity flying chips.



Handling Metal Stock

Hazard: Sharp edges, burrs, burns.

NEVER wear gloves on rotating machinery (drill press/lathe) – Entanglement Hazard!



General Floor

Hazard: Drop hazards, flame.

WORKSHOP CONTROLS

Housekeeping

5S Principles: Keep walkways clear of swarf/oil.



Machine Guarding

Ensure interlocks intact over belts/gears before power-up.



Isolation (Lockout/Tagout)

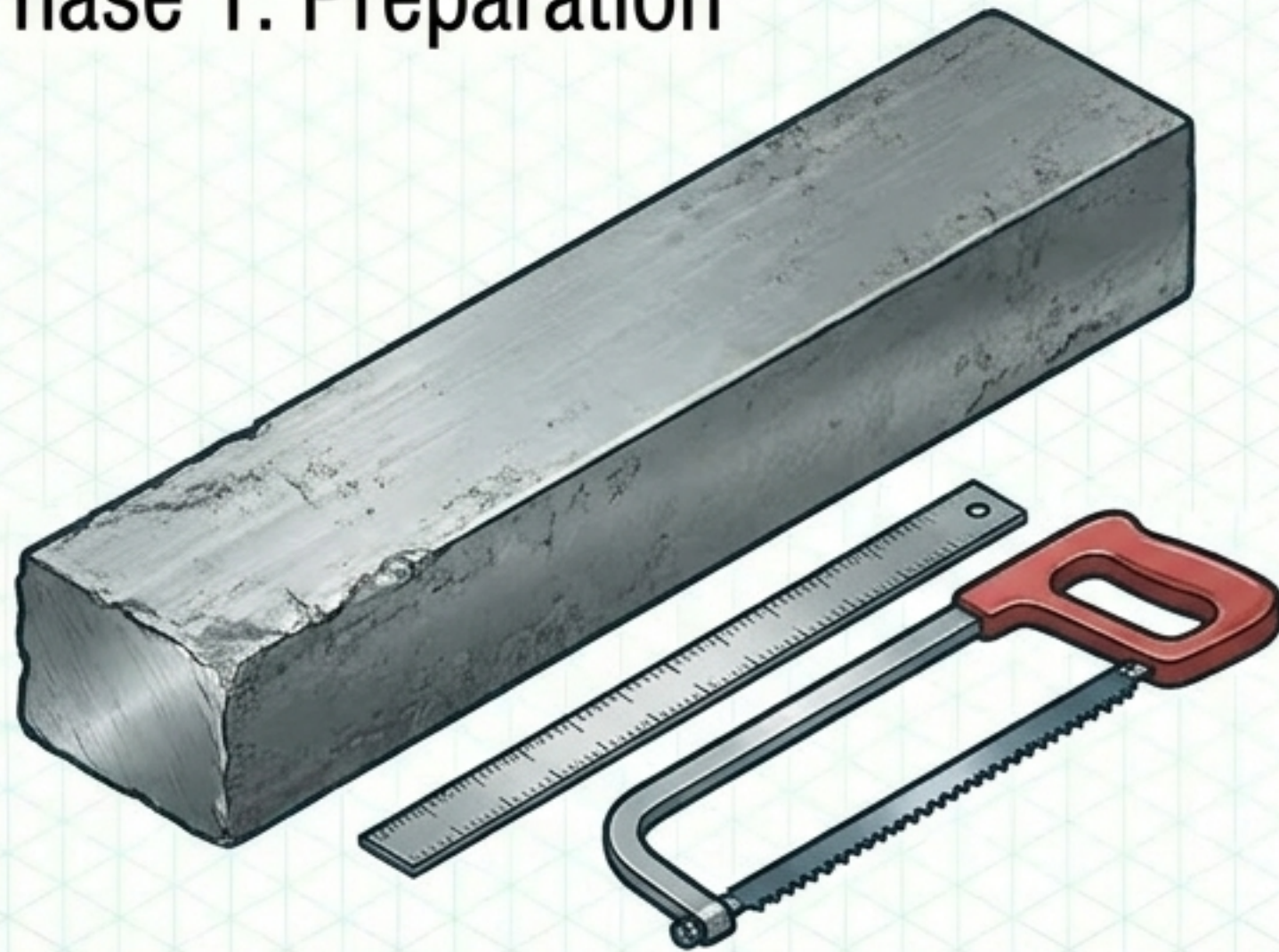


Disconnect primary power and lock switchboard before tool changes.



Practical Case Study: Manufacturing the G-Strap (Part 1 of 3)

Phase 1: Preparation

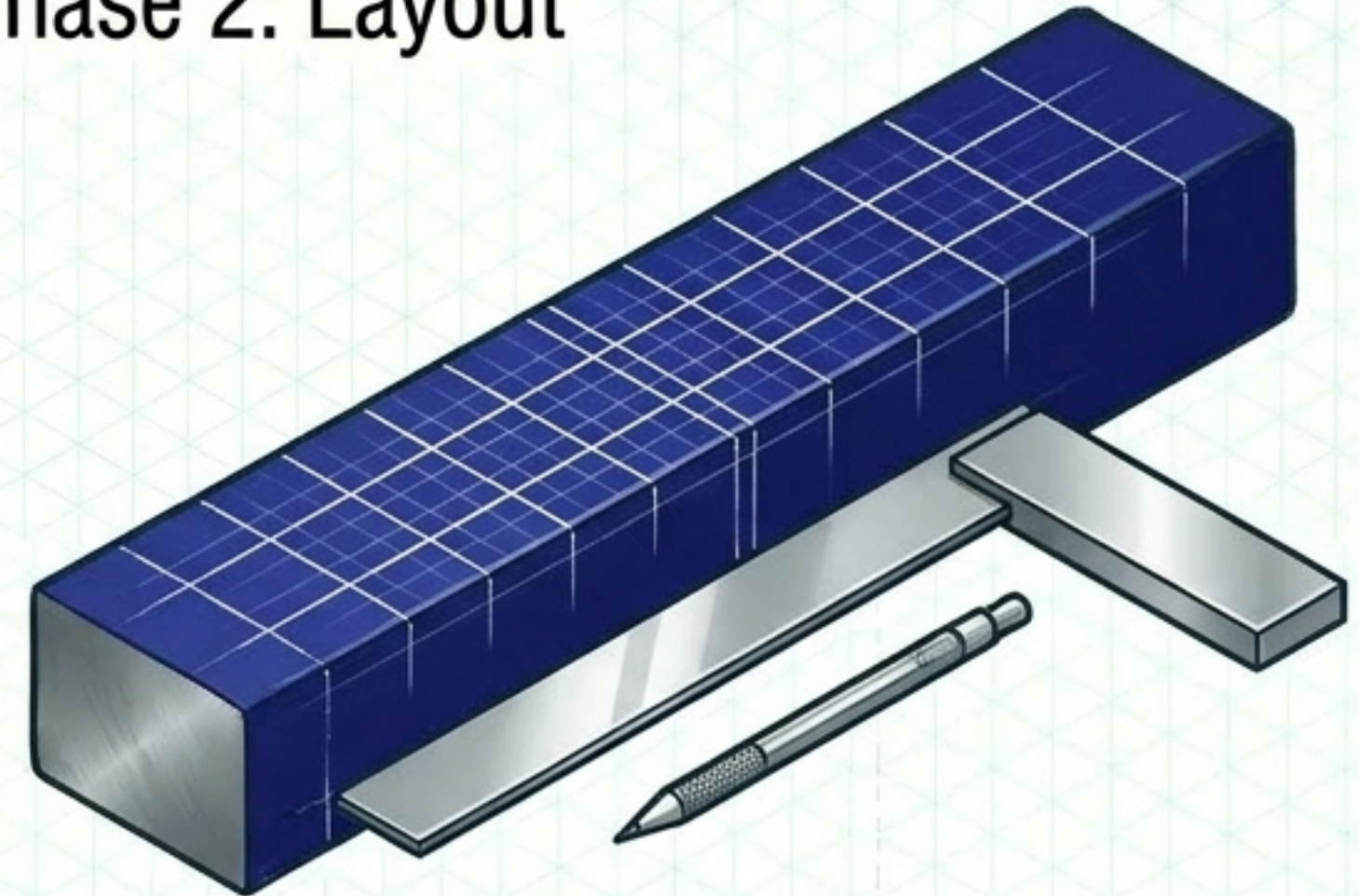


Operation: Material Inspection & Rough Cutting

Tool Check: Hacksaw (18 TPI selected for thick material).

Critical Metric: Inspect defects, allow machining allowance.

Phase 2: Layout



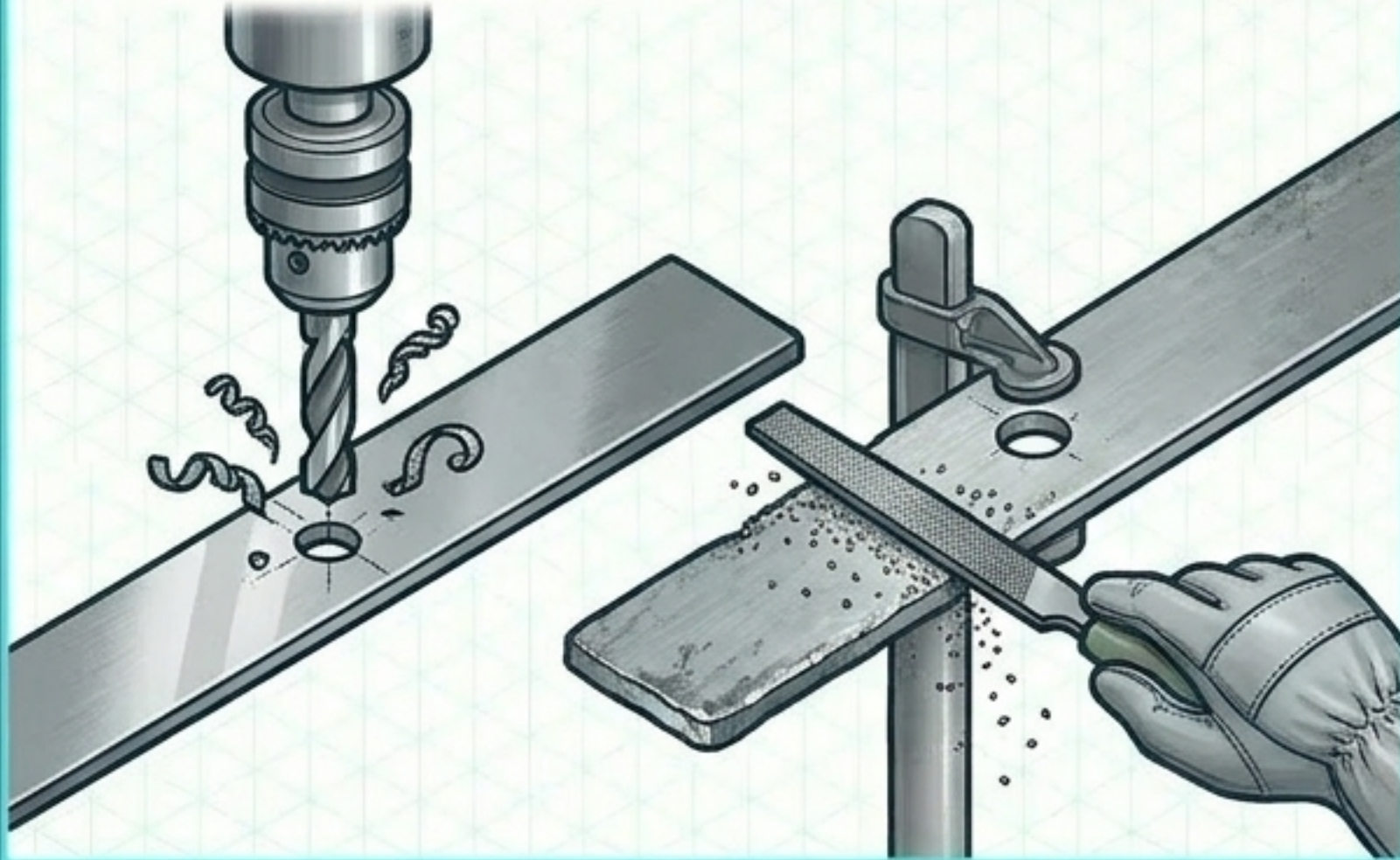
Operation: Marking Out Coordinates

Tool Check: Engineering Blue, Scribe, Try Square, Prick Punch (30°).

Critical Metric: Establish datum edge; ensure exactly 90° intersections.

Practical Case Study: Manufacturing the G-Strap (Part 2 of 3)

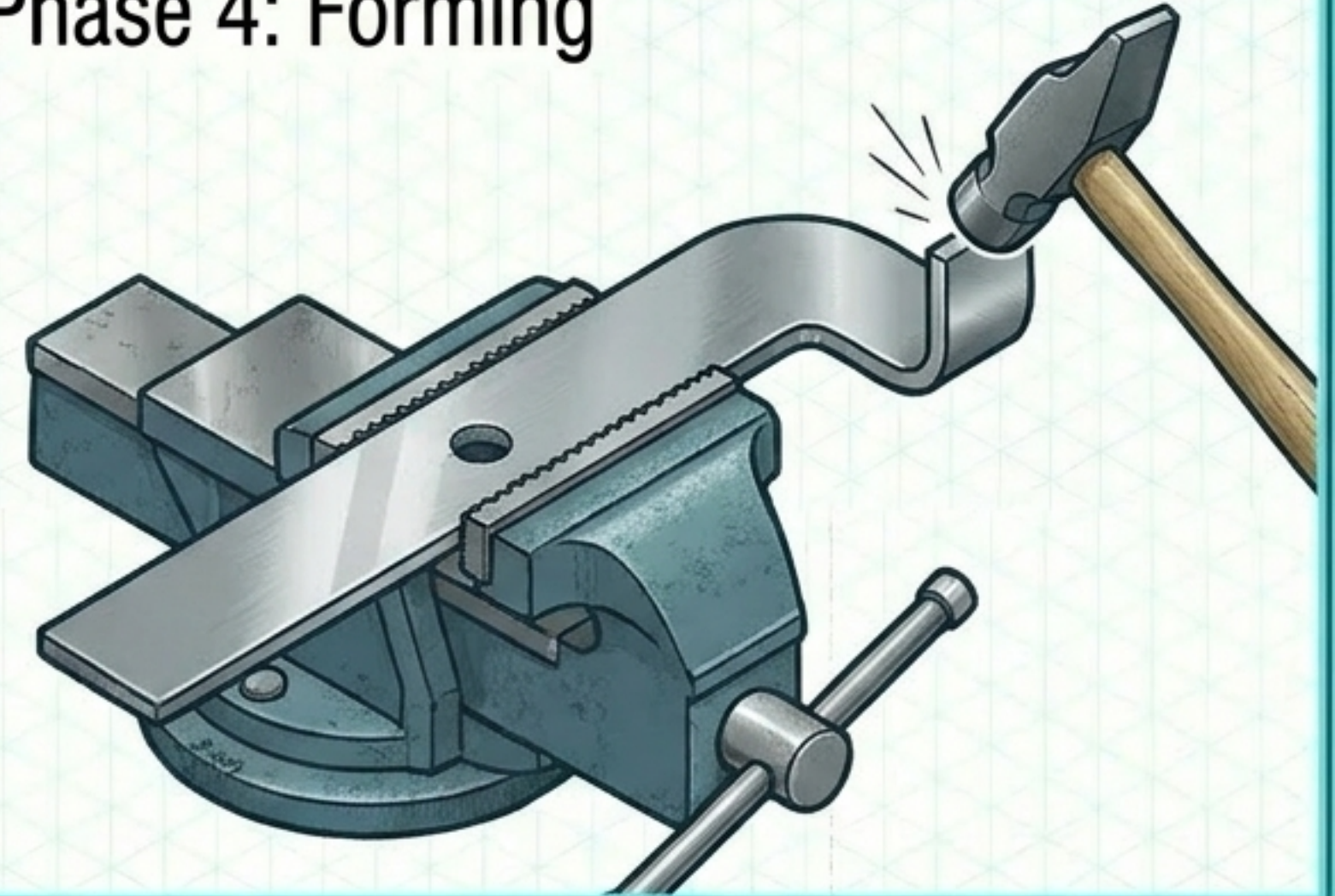
Phase 3: Removal



Operation: Hole Prep & Perimeter Finishing

Tool Check: Center Punch (90°), Drill Press, Bastard/Second-Cut Files.
Critical Metric: Center punch deeply to prevent drill walk. File edges to remove burrs for safe handling.

Phase 4: Forming

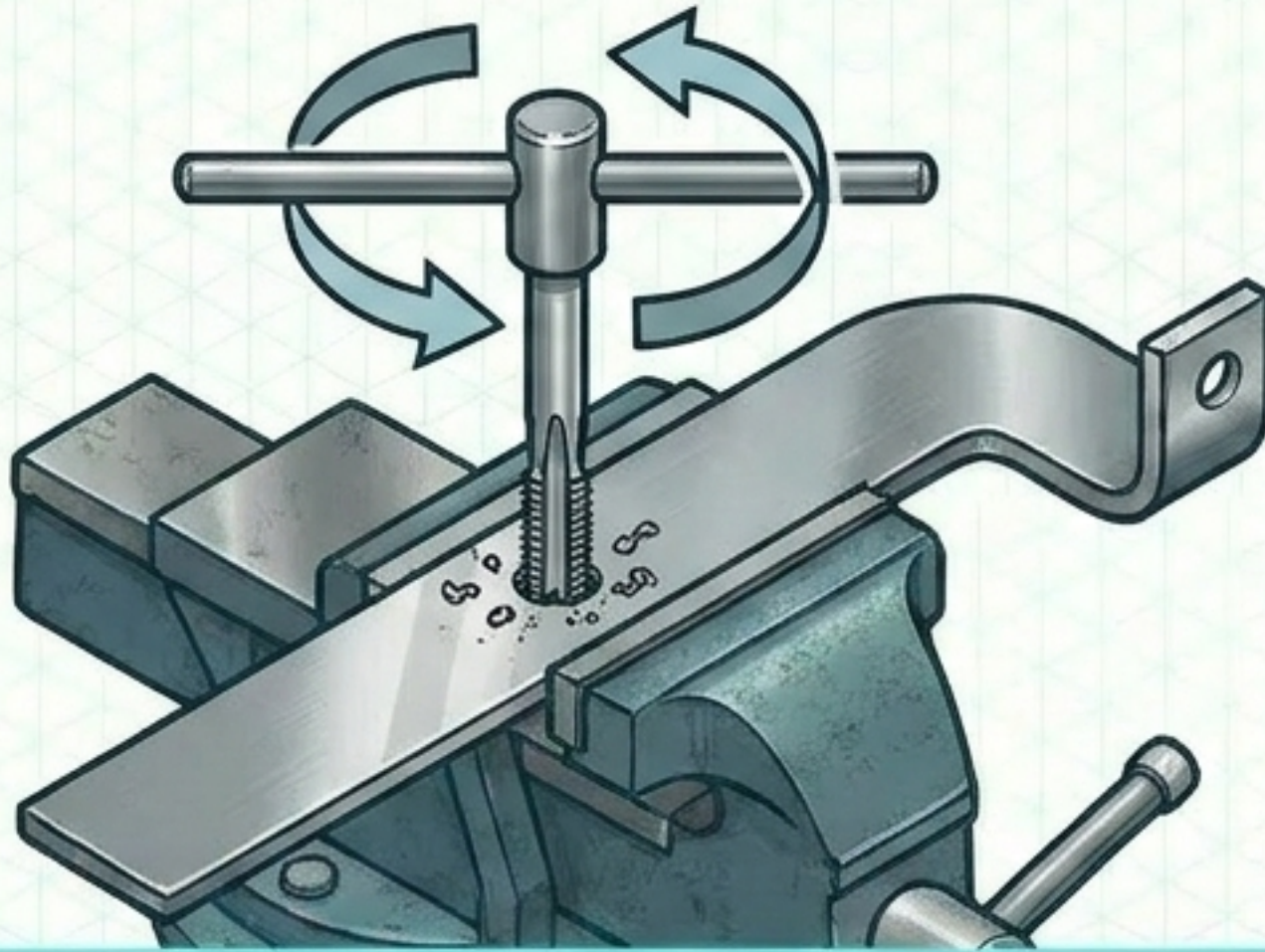


Operation: Bending the Profile

Tool Check: Bench Vice, Hammer / Press.
Critical Metric: Account for springback by over-bending slightly past target angle.

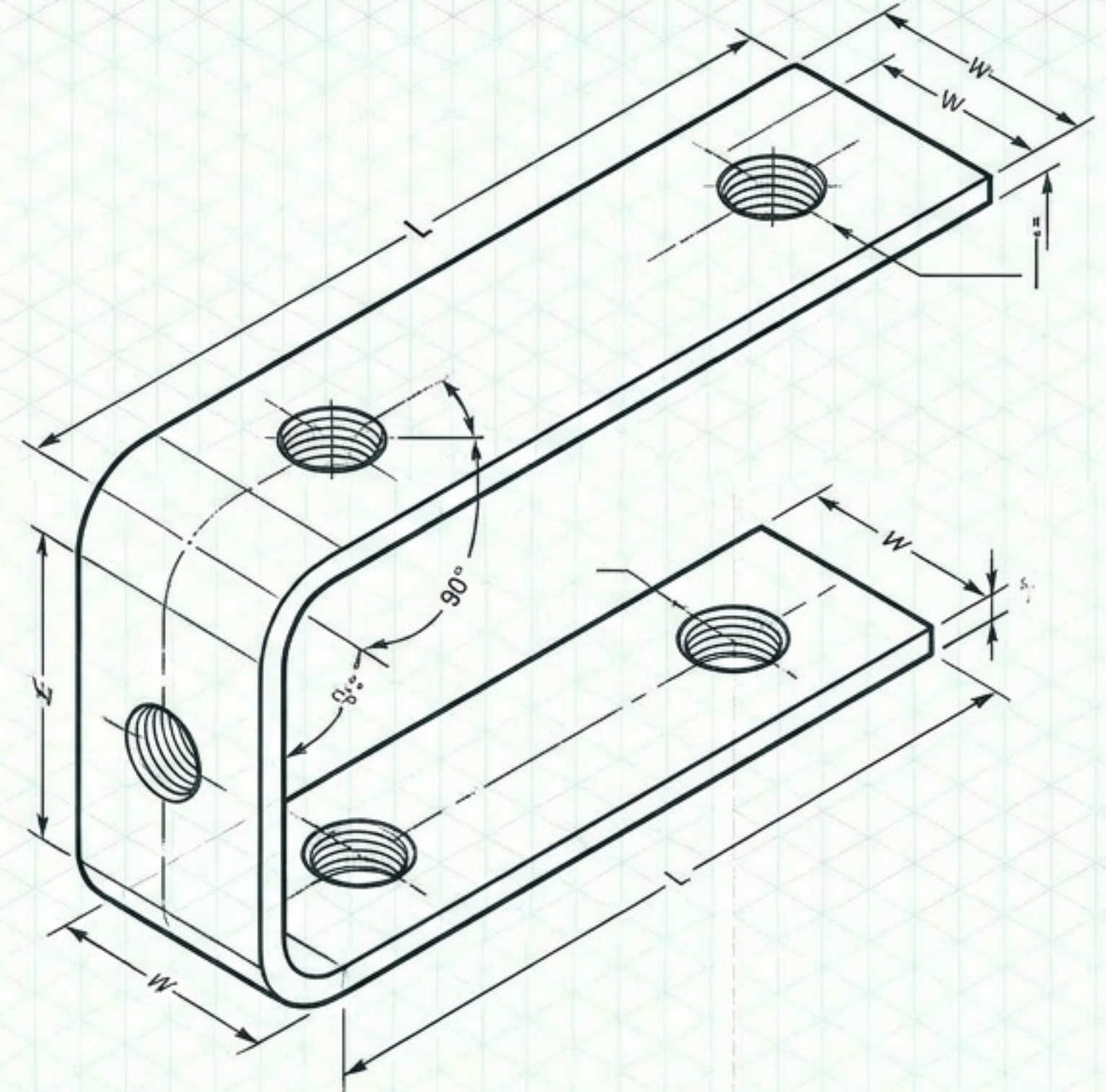
Practical Case Study: Manufacturing the G-Strap (Part 3 of 3)

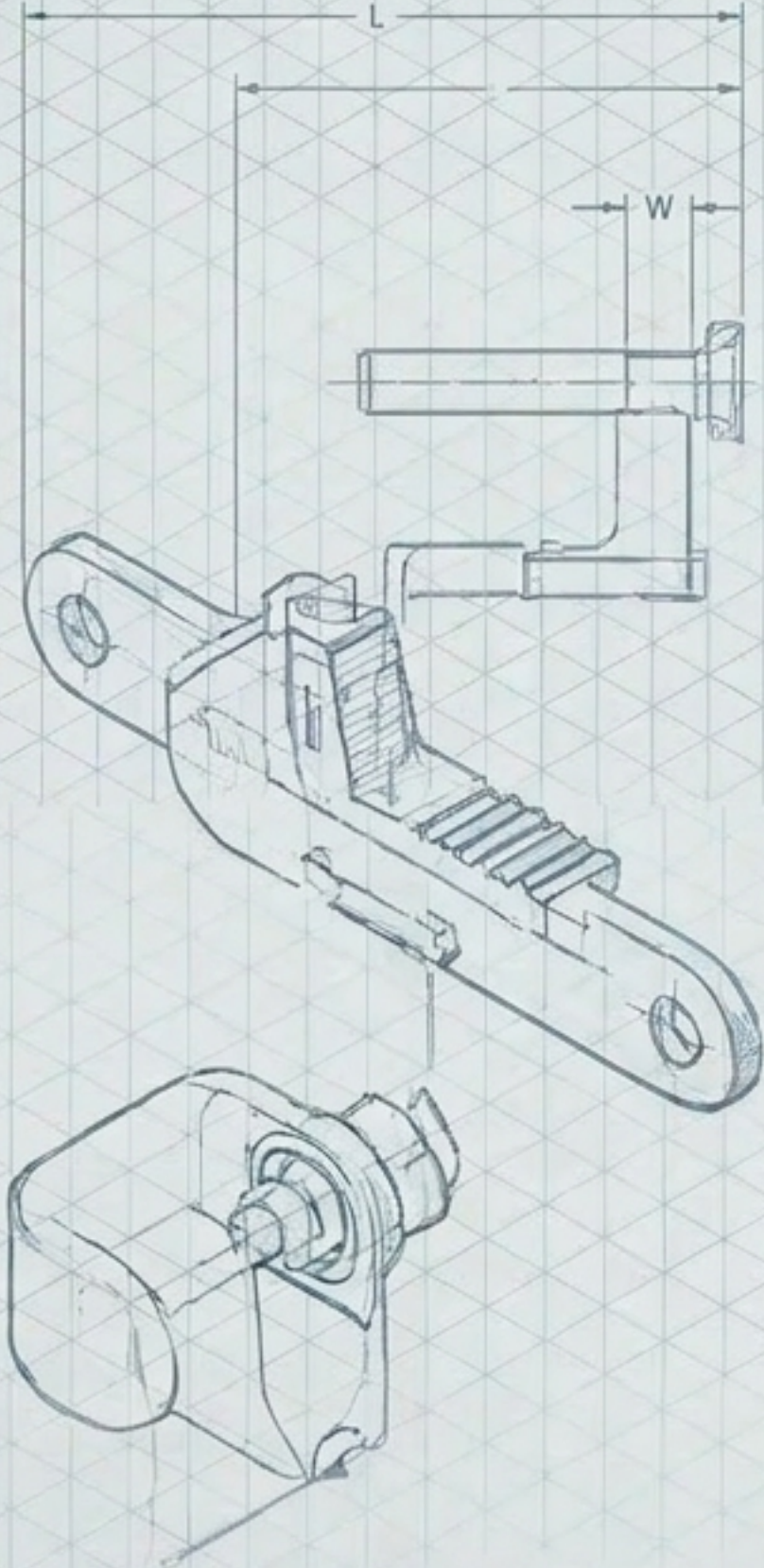
Phase 5: Threading



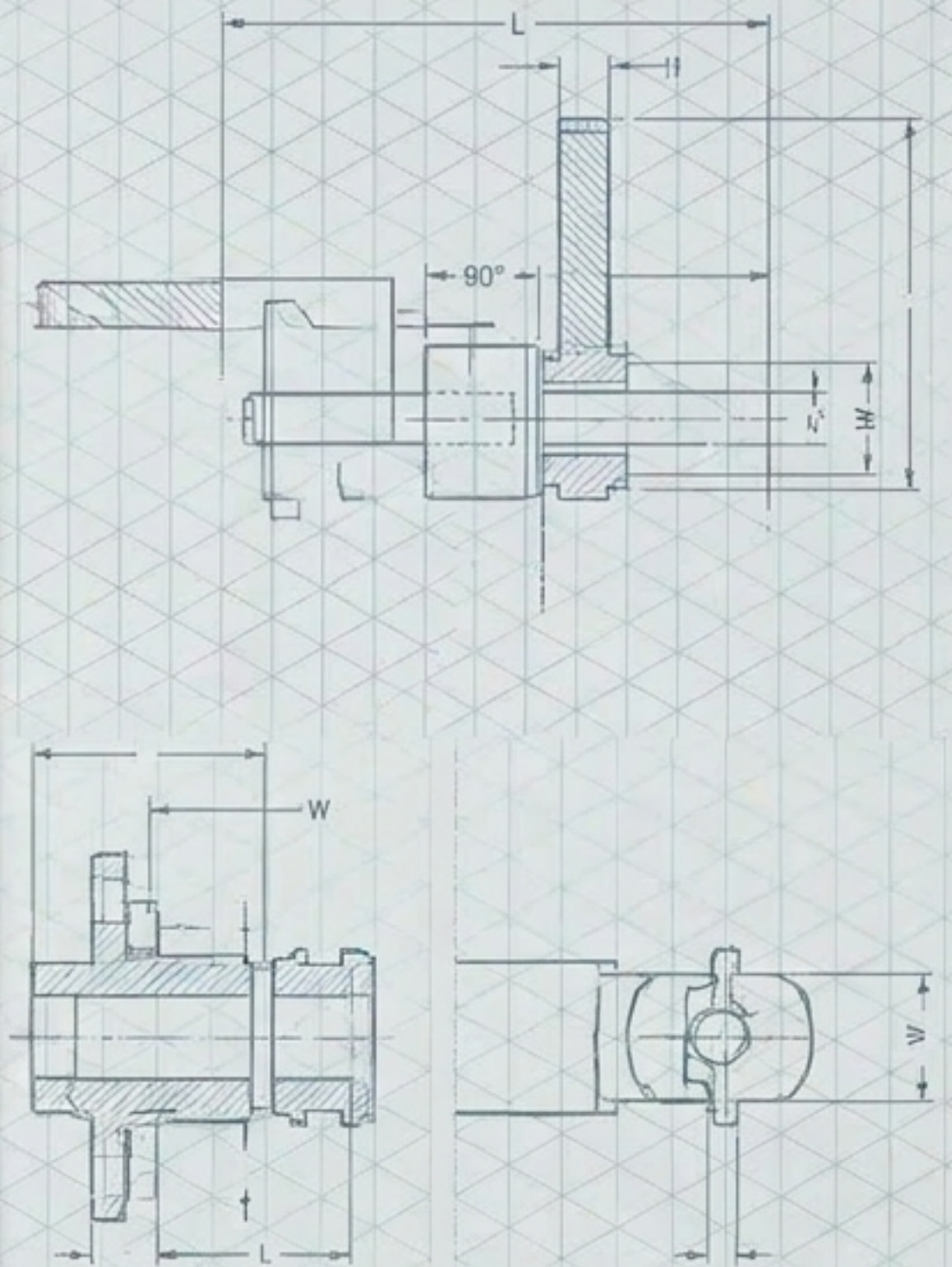
Operation: Cutting Internal Clamping Threads

Tool Check: Taper, Intermediate, and Bottoming Hand Taps.
Critical Metric: Calculate correct Tap Drill Size ($TDS = D - P$).
Remember to reverse turn to break chips.





- ✓ **Metrology:** Vernier & Micrometer verified.
- ✓ **Marking:** 30° Prick & 90° Center punches aligned.
- ✓ **Cutting:** TPI snag rules applied.
- ✓ **Forming:** Springback and Neutral Axis calculated.
- ✓ **Threading:** TDS computed; 3-Tap Sequence executed.
- ✓ **Safety:** PPE isolated and Guarding confirmed.



8-Hour Blueprint Complete. Proceed to Assessment.